APPENDIX D: CONVEX APPROXIMATIONS FOR UAV TRANSMIT-POWER OPTIMIZATION

In this appendix, we provide the convex approximations for the UAV transmit-power optimization problem (P4). Define

$$\omega[n] \triangleq \frac{\delta - \tau[n]}{\delta}, \qquad a_u[n] \triangleq \sum_{b\in\mathcal{B}} \mu_{u,b}[n]. \tag{116}$$

The legitimate spectral efficiency of UAV u can be expressed in difference-of-concave form as

$$R_u^{\mathrm{leg}}[n] = \sum_{b\in\mathcal{B}} \mu_{u,b}[n] \left(F_{u,b}^{(1)}[n] - F_{u,b}^{(2)}[n] \right), \tag{117}$$

where

$$F_{u,b}^{(1)}[n] = \log_2 \left(\sigma_b^2 + \sum_{i\in\mathcal{U}} p_i[n] h_{i,b}[n] \right), \tag{118}$$

$$F_{u,b}^{(2)}[n] = \log_2 \left(\sigma_b^2 + \sum_{\substack{i\in\mathcal{U} \\ i\neq u}} p_i[n] h_{i,b}[n] \right). \tag{119}$$

At the local power point $\mathbf{P}^{(\ell)}$, the concavity of $F_{u,b}^{(2)}[n]$ yields the global affine upper bound

$$\begin{aligned} \overline{F}_{u,b}^{(2,\ell)}[n] =& F_{u,b}^{(2)}\left(\mathbf{P}^{(\ell)}\right) \\ &+ \sum_{\substack{i\in\mathcal{U} \\ i\neq u}} \frac{h_{i,b}[n] \left(p_i[n] - p_i^{(\ell)}[n] \right)}{\ln 2 \, \Psi_{u,b}^{(\ell)}[n]}, \end{aligned} \tag{120}$$

where

$$\Psi_{u,b}^{(\ell)}[n] = \sigma_b^2 + \sum_{\substack{i\in\mathcal{U} \\ i\neq u}} p_i^{(\ell)}[n] h_{i,b}[n]. \tag{121}$$

Consequently, a concave lower approximation of the legitimate spectral efficiency is

$$\underline{R}_u^{\mathrm{leg},(\ell)}[n] = \sum_{b\in\mathcal{B}} \mu_{u,b}[n] \left(F_{u,b}^{(1)}[n] - \overline{F}_{u,b}^{(2,\ell)}[n] \right). \tag{122}$$

For Eve e, let

$$I_e^{\mathrm{AN}}[n] \triangleq \sum_{b\in\mathcal{B}} \Phi_{b,e}^{\mathrm{low}}[n] \tag{123}$$

denote the minimum AN interference guaranteed by all BSs. The worst-case eavesdropping spectral efficiency is written as

$$R_{u,e}^{\mathrm{up}}[n] = a_u[n] \left(G_{u,e}^{(1)}[n] - G_{u,e}^{(2)}[n] \right), \tag{124}$$

where

$$G_{u,e}^{(1)}[n] = \log_2 \left(\mathcal{I}_{u,e}^{\mathrm{low}}[n] + p_u[n] h_{u,e}^{\mathrm{up}}[n] \right), \tag{125}$$

$$G_{u,e}^{(2)}[n] = \log_2 \left(\mathcal{I}_{u,e}^{\mathrm{low}}[n] \right), \tag{126}$$

and

$$\mathcal{I}_{u,e}^{\mathrm{low}}[n] = I_e^{\mathrm{AN}}[n] + \sigma_e^2 + \sum_{\substack{i\in\mathcal{U}\\ i\neq u}} p_i[n] h_{i,e}^{\mathrm{low}}[n]. \tag{127}$$

Since $G_{u,e}^{(1)}[n]$ is concave in $\mathbf{P}$, its first-order expansion provides a global affine upper bound:

$$\begin{aligned}\overline{G}_{u,e}^{(1,\ell)}[n] =& G_{u,e}^{(1)}\left(\mathbf{P}^{(\ell)}\right) \\ &+ \frac{h_{u,e}^{\mathrm{up}}[n]\left(p_u[n] - p_u^{(\ell)}[n]\right)}{\ln 2\, \Lambda_{u,e}^{(\ell)}[n]} \\ &+ \sum_{\substack{i\in\mathcal{U}\\ i\neq u}} \frac{h_{i,e}^{\mathrm{low}}[n]\left(p_i[n] - p_i^{(\ell)}[n]\right)}{\ln 2\, \Lambda_{u,e}^{(\ell)}[n]},\end{aligned} \tag{128}$$

where

$$\Lambda_{u,e}^{(\ell)}[n] = I_e^{\mathrm{AN}}[n] + \sigma_e^2 + p_u^{(\ell)}[n] h_{u,e}^{\mathrm{up}}[n] + \sum_{\substack{i\in\mathcal{U}\\ i\neq u}} p_i^{(\ell)}[n] h_{i,e}^{\mathrm{low}}[n]. \tag{129}$$

Therefore, a convex upper approximation of the worst-case eavesdropping rate is

$$\overline{R}_{u,e}^{\mathrm{up},(\ell)}[n] = a_u[n]\left(\overline{G}_{u,e}^{(1,\ell)}[n] - G_{u,e}^{(2)}[n]\right). \tag{130}$$